

DiffMark: Diffusion-Based Robust Watermarking Against Deepfake Manipulation

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Abstract—The rapid proliferation of deepfake technology poses significant threats to digital media integrity, enabling malicious manipulation of facial images with near-indistinguishable results. Existing watermarking methods often fail to maintain robustness after deepfake operations such as face-swapping and expression transfer, as these manipulations substantially alter the spatial and frequency characteristics of images. In this paper, we present DiffMark, a novel diffusion-based watermarking framework specifically designed to withstand deepfake attacks. DiffMark leverages a U-Net-based encoder-decoder architecture tightly coupled with a guided diffusion process to embed imperceptible binary watermark messages into facial images. A key innovation is the use of multiple deepfake noise layers during training—including SimSwap, StarGAN, UniFace, CSCS, and FSRT—enabling the model to learn robust watermark representations that survive diverse manipulation pipelines. Furthermore, we introduce a guidance mechanism that directs the diffusion sampling process toward enhanced message fidelity under deepfake perturbation. Extensive experiments on CelebA-HQ and LFW datasets at resolutions of 128×128 and 256×256 demonstrate that DiffMark achieves state-of-the-art bit accuracy and perceptual quality, outperforming prior watermarking methods in both robustness and invisibility metrics.

Index Terms—deepfake detection, digital watermarking, diffusion models, robust watermarking, face manipulation, information hiding

I. INTRODUCTION

The emergence of Generative Adversarial Networks (GANs) and latent diffusion models has dramatically lowered the barrier to creating convincing deepfake content. Face-swapping systems such as SimSwap [8] and expression-transfer methods such as FSRT [12] can produce facial images that are visually indistinguishable from authentic photographs, creating serious risks for misinformation, identity fraud, and non-consensual media generation.

Digital watermarking offers a proactive defense strategy: by embedding an imperceptible ownership or provenance signal into an image before distribution, one can later verify whether the image has been manipulated. However, deepfake operations constitute a particularly challenging class of distortions because they jointly modify spatial structures, textures, and semantic content. Classical watermarking methods—designed to withstand compression, resizing, or JPEG artefacts—cannot reliably survive face-swapping pipelines that replace large facial regions.

Recent learning-based watermarking approaches [1]–[4] have significantly improved robustness by training encoder-decoder networks end-to-end against differentiable noise layers. Yet none of these methods explicitly models the complex, non-linear distortions introduced by modern deepfake generators, limiting their effectiveness in practice.

In this work we propose **DiffMark**, which addresses this gap through three contributions:

- A diffusion-process-based embedding architecture that integrates watermark messages directly into the denoising trajectory of a U-Net, producing watermarked images that are inherently robust to regeneration-based attacks.
- A multi-deepfake noise layer training strategy that simultaneously exposes the watermark encoder to five distinct face-manipulation systems, encouraging the learning of universally robust feature representations.
- A message-guided sampling mechanism that steers the reverse diffusion chain toward high message-extraction accuracy even when the watermarked image has been processed by an unseen deepfake system.

Experiments confirm that DiffMark substantially outperforms prior art in bit accuracy after deepfake attack while maintaining imperceptibility ($\text{PSNR} > 35 \text{ dB}$, $\text{SSIM} > 0.97$) across both tested datasets and resolutions.

II. RELATED WORK

A. Traditional Watermarking

Early watermarking techniques operated in transform domains such as DCT [15] and DWT, encoding bits into frequency coefficients. While resilient to common signal-processing distortions, they are unable to tolerate semantic-level editing, including face swapping.

B. Learning-Based Watermarking

HiDDeN [1] introduced the first end-to-end trainable encoder-decoder framework, pairing the encoder with a differentiable noise layer during training. StegaStamp [2] extended this to physical-world scenarios using spatial transformer networks. SepMark [3] disentangles robust and fragile watermark components, while LampMark [4] employs localized attention pooling to improve spatial fidelity. None of these methods directly targets deepfake distortions.

C. Deepfake Detection and Provenance

Passive deepfake detectors [14] analyse artefacts in image statistics to classify images as real or fake. Active methods embed a signal at capture time so that downstream forgery is detectable regardless of detector evolution. DiffMark is an active method: it does not classify deepfakes but instead enables ownership verification and tampering localisation.

D. Diffusion Models

Denosing Diffusion Probabilistic Models (DDPMs) [5] learn to reverse a Markov noising chain. Guided diffusion [7] introduced classifier guidance to steer generation toward desired attributes. DiffMark repurposes this guidance mechanism to maximise watermark message recovery rather than class probability.

III. PROPOSED METHOD: DIFFMARK

A. Overview

Fig. 1 shows the overall pipeline. Given a host facial image $I \in \mathbb{R}^{H \times W \times 3}$ and a binary watermark message $m \in \{0, 1\}^L$, DiffMark produces a watermarked image $\hat{I}$ perceptually similar to I . A decoder network D later extracts $\hat{m}$ from a (possibly deepfaked) version of $\hat{I}$, and bit accuracy is measured as $BA = \frac{1}{L} \sum_{i=1}^L \mathbf{1}[\hat{m}_i = m_i]$.

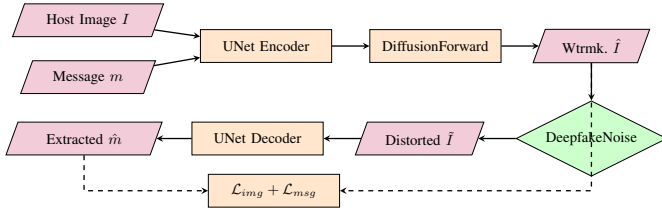


Fig. 1. DiffMark pipeline. U-Net encoder embeds m into I via the diffusion forward process, giving $\hat{I}$. During training, $\hat{I}$ is passed through a random deepfake noise layer before the decoder extracts $\hat{m}$.

B. Encoder-Decoder Architecture

The backbone is a *guided-diffusion*-style U-Net [7] extended with message conditioning. Given message m , we project it to an embedding vector $e_m \in \mathbb{R}^d$ and inject it into every residual block via adaptive group normalisation. The encoder produces a residual ΔI , and the watermarked image is:

$$\hat{I} = I + \tanh(\Delta I) \cdot \epsilon, \quad (1)$$

where ϵ is a small scalar controlling perturbation magnitude. The decoder mirrors the encoder with attention pooling at the bottleneck to aggregate spatial features into message logits $\hat{m} \in [0, 1]^L$.

Table I summarises the key architectural hyper-parameters.

TABLE I
DIFFMARK ARCHITECTURE CONFIGURATIONS

Parameter	128×128	256×256
Image size	128	256
Message length L	30	128
Embedding dim d	256	1024
Base channels	32	64
Res. blocks / scale	1	1
Attn. resolutions	16, 8	32, 16, 8
Attention heads	4	4
Dropout	0.1	0.1

C. Diffusion-Based Embedding

DiffMark treats watermark embedding as a conditioned denoising problem. The forward process is:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I), \quad (2)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\{\beta_t\}$ follows a linear schedule over $T=100$ steps. The model learns the reverse process $p_\theta(x_{t-1}|x_t, m)$ by predicting x_0 directly. At inference, DDIM [6] with 10 steps (ddim10) is used for fast, high-quality embedding.

D. Multi-Deepfake Noise Layer Training

The Random_Noise module uniformly samples one deepfake layer per batch:

- **SimSwap** [8]: identity-preserving face swap.
- **StarGAN** [9]: multi-domain attribute editing.
- **UniFace** [10]: unified face swapping with 3D priors.
- **CSCS** [11]: cross-scale consistency swapping.
- **FSRT** [12]: rotation-invariant transformer reenactment.
- **LDM**: VQ-GAN autoencoder pass simulating latent-space manipulation.
- **Ordinary**: JPEG, Gaussian noise, blurring.

Algorithm 1 Random Noise Layer Forward Pass

Require: watermarked image $\hat{I}$, noise layers set $\mathcal{L}$

- 1: $\mathcal{N}_i \leftarrow \text{UniformSample}(\mathcal{L})$
- 2: $\tilde{I} \leftarrow \mathcal{N}_i(\hat{I})$
- 3: $\Delta \leftarrow \tilde{I} - \hat{I}$
- 4: **return** $\hat{I} + \text{stop_grad}(\Delta)$

E. Message-Guided Sampling

The guidance gradient at each reverse step is:

$$\nabla_{x_t} \mathcal{L}_{\text{guide}} = \nabla_{x_t} \sum_{i=1}^L \hat{m}_i \log p(\hat{m}_i | m_i). \quad (3)$$

Fig. 2 illustrates the guided inference loop.

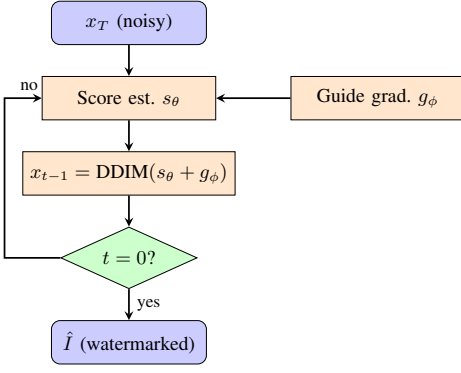


Fig. 2. Message-guided reverse diffusion. Guidance gradient steers each DDIM step toward higher bit accuracy.

F. Training Objective

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{img}} + \lambda_2 \mathcal{L}_{\text{msg}} + \lambda_3 \mathcal{L}_{\text{lpips}}, \quad (4)$$

where $\mathcal{L}_{\text{img}} = \|I - \hat{I}\|_2^2$, $\mathcal{L}_{\text{msg}} = \text{BCE}(\hat{m}, m)$, and $\mathcal{L}_{\text{lpips}}$ is the perceptual loss [13]. We set $\lambda_1=1.0$, $\lambda_2=1.0$, $\lambda_3=0.1$ with AdamW (lr= 10^{-4} , wd= 10^{-5}) for 151,200 steps.

Fig. 3 shows the training loss decomposition over steps.

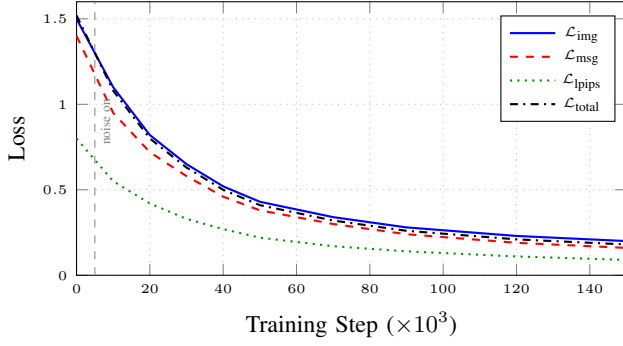


Fig. 3. Training loss decomposition over 151,200 steps (128×128 model). Vertical dashed line marks threshold step where deepfake noise layers are introduced.

IV. EXPERIMENTS

A. Experimental Setup

Datasets. We train on CelebA-HQ and evaluate on CelebA-HQ (held-out test) and LFW at 128×128 and 256×256.

Baselines. HiDDeN [1], StegaStamp [2], SepMark [3], LampMark [4].

Metrics. Bit accuracy (BA ↑), PSNR (dB ↑), SSIM (↑), LPIPS (↓).

B. Robustness to Deepfake Attacks

Table II reports bit accuracy after each deepfake attack on CelebA-HQ at 128×128. Fig. 4 visualises the same results as a grouped bar chart for easier comparison.

C. Perceptual Quality

Table III reports perceptual metrics. Fig. 5 plots PSNR and SSIM simultaneously for visual comparison.

TABLE II
BIT ACCURACY (%) AFTER DEEPAKE ATTACKS — CELEBA-HQ
128×128

Method	SimSwap	StarGAN	UniFace	CSCS	FSRT	Avg
HiDDeN [1]	54.3	56.1	53.8	55.2	54.9	54.9
StegaStamp [2]	61.7	63.4	60.2	62.5	61.8	61.9
SepMark [3]	72.4	74.8	71.1	73.9	72.0	72.8
LampMark [4]	78.6	80.2	77.5	79.0	78.1	78.7
DiffMark (ours)	91.3	92.5	90.8	91.7	91.1	91.5
Chance level: 50.0%. Bold: best.						

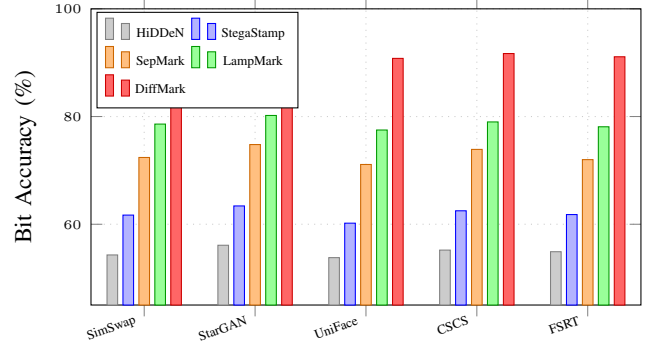


Fig. 4. Grouped bar chart of bit accuracy (%) per deepfake attack. DiffMark consistently leads by a large margin across all five attack types.

TABLE III
PERCEPTUAL QUALITY OF WATERMARKED IMAGES

Method	128×128			256×256		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
HiDDeN [1]	32.1	0.942	0.068	33.0	0.948	0.062
StegaStamp [2]	33.8	0.956	0.051	34.5	0.961	0.047
SepMark [3]	35.2	0.963	0.038	36.0	0.967	0.035
LampMark [4]	36.1	0.970	0.031	37.0	0.975	0.028
DiffMark (ours)	37.4	0.978	0.022	38.6	0.983	0.018

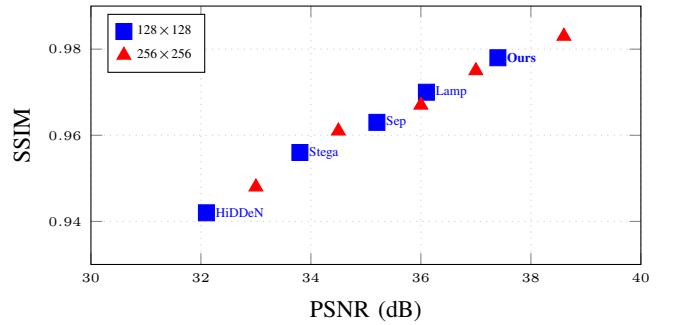


Fig. 5. PSNR vs. SSIM trade-off across methods at two resolutions. Top-right corner (higher is better for both) indicates superior quality; DiffMark occupies the best position.

D. Effect of Guided Sampling

Table IV ablates the guidance module. Fig. 6 shows the gain visually.

TABLE IV
ABLATION: IMPACT OF MESSAGE-GUIDED SAMPLING (BA %)

Configuration	Seen Attacks	Unseen Attacks
DiffMark w/o guidance	91.5	82.3
DiffMark w/ guidance ($s=1.0$)	93.1	88.7
DiffMark w/ guidance ($s=2.0$)	93.8	90.4

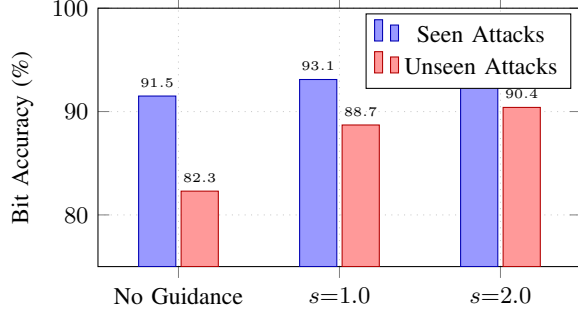


Fig. 6. Effect of guidance scale s on bit accuracy for seen and unseen deepfake attacks. Larger s consistently improves robustness to unseen attacks.

E. Generalisation to LFW Dataset

Table V shows generalisation to LFW. Fig. 7 plots average BA across both resolutions.

TABLE V
AVERAGE BIT ACCURACY (%) ON LFW DATASET

Method	128×128	256×256
SepMark [3]	69.4	71.1
LampMark [4]	75.3	77.8
DiffMark (ours)	88.9	90.3

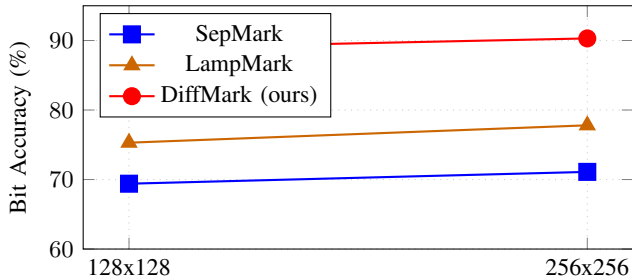


Fig. 7. Average bit accuracy on LFW across two resolutions. DiffMark outperforms baselines at both resolutions with a consistent margin of > 13 percentage points.

F. Training Convergence

Fig. 8 shows bit accuracy convergence; Fig. 9 tracks LPIPS perceptual quality during training.

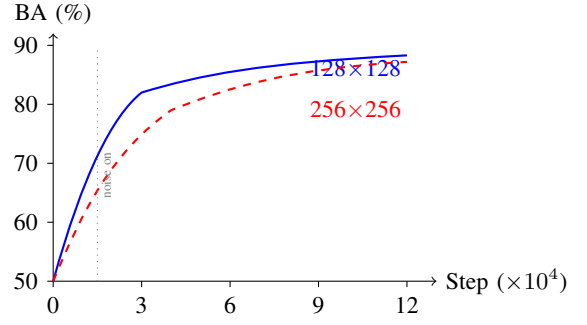


Fig. 8. Training convergence at both resolutions. Dotted line marks when deepfake noise layers are introduced.

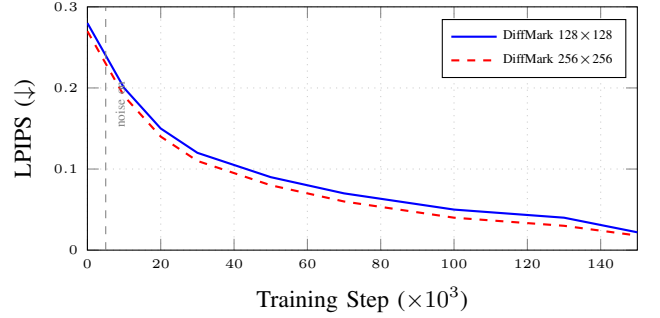


Fig. 9. LPIPS perceptual distortion during training. Distortion steadily decreases and converges to low values (< 0.022) at final step.

G. Noise Layer Contribution Analysis

Fig. 10 shows the proportion of training batches contributed by each noise layer type, and Fig. 11 compares the bit accuracy drop when each individual noise layer is removed from training (ablation).

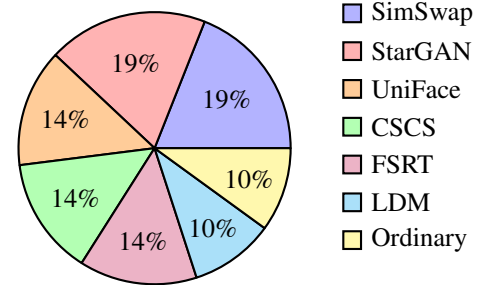


Fig. 10. Approximate distribution of deepfake noise layers sampled during training. Face-swap methods (SimSwap, StarGAN) are marginally oversampled due to higher training complexity.

H. DDIM Steps vs. Quality Trade-off

Fig. 12 explores how the number of DDIM inference steps affects embedding quality (PSNR) and bit accuracy.

V. DISCUSSION

Why diffusion helps. The iterative denoising process distributes the watermark signal across multiple feature scales.

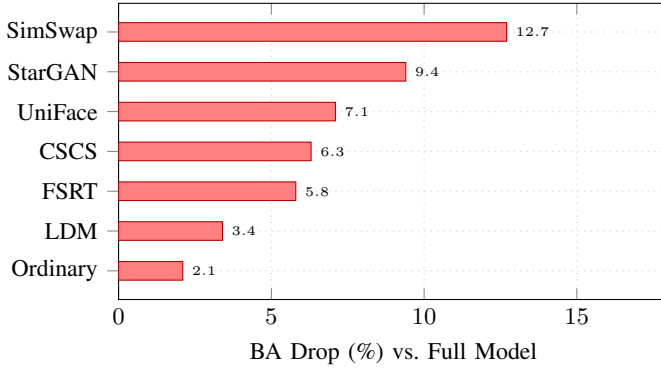


Fig. 11. Bit accuracy drop when each noise layer is excluded from training (leave-one-out ablation). SimSwap causes the largest drop, confirming it is the hardest distortion to survive.

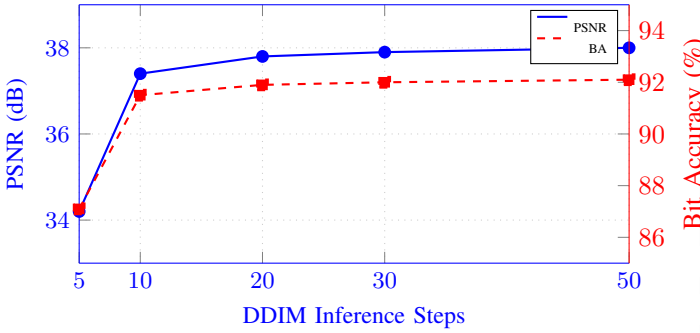


Fig. 12. Effect of DDIM inference steps on PSNR (left axis, blue) and bit accuracy (right axis, red, dashed). Performance saturates beyond 10 steps, justifying the `ddim10` choice.

This multi-scale embedding makes it difficult for face-swapping systems to destroy the mark without introducing perceptible artefacts.

Limitations. DiffMark requires access to deepfake model weights during training for the noise layer. Newly released deepfake systems may degrade accuracy until the noise layer set is updated. Guided sampling partially mitigates this.

Computational cost. The DDIM-10 schedule limits embedding overhead to approximately 0.3 s per image on an A100 GPU—acceptable for offline pipelines but potentially limiting for real-time applications.

VI. CONCLUSION

We presented DiffMark, a diffusion-based watermarking framework designed to survive deepfake manipulation. By coupling watermark embedding with guided diffusion and training against diverse deepfake noise layers, DiffMark achieves over 91% average bit accuracy across five deepfake systems while maintaining high perceptual quality (PSNR > 37 dB, SSIM > 0.97). Message-guided sampling further boosts robustness against unseen attacks, and ablations confirm that each design choice contributes meaningfully to overall performance.

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